PM Graded Quiz M4

Question 1

Kahneman and Tversky use the following weighting function:

$$w^+(p) = \frac{p^\gamma}{[p^\gamma + (1-p)^\gamma]^{1/\gamma}}$$

for the probability of outcomes that lead to gains, and

$$w^-(p) = \frac{p^\delta}{[p^\delta + (1-p)^\delta]^{1/\delta}}$$

for the probability of outcomes that lead to losses instead.

What happens to $w^+(p)$ if $\delta \to 1$?

- ◯ $w^+(p) \to \ln(p)$ or individuals consider low probabilities bigger than they actually are and large probabilities as smaller than they are.
- ◯ $w^+(p) \to 0$ so the individuals discounts both high and low probabilities as there was no probability.
- ◯ $w^p = 1$ i.e., all events are considered as though they are certain outcomes.
- ◯ $w^+(p) \to p$, the perfectly risk neutral weighting.

Question 2

In contrast to traditional finance, what does the empirical evidence collected by Kahneman and Tversky say about the behavior of individuals on average when faced with the following gambling situation in which "they have an equal chance to win $100 or lose $70"?

- ◯ They do not take the gamble despite its expected value being positive.
- ◯ They do take the gamble as the expected value is positive.
- ◯ They do not take the gamble because the expected value is negative.
- ◯ They do take the gamble even though its expected value is negative because they are risk seekers.

Question 3

In applying BMAA, it is very important to be able to detect clients' behavioral biases. Associate the correct behavioral bias to a suitable question among those in the list below.

A) “In those occasions where you invested money in stocks based on your own research, when there were indications in the news that those companies were not doing well or that the economy was not as strong as you

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predicted, did you choose to ignore those information or discount them compared to your own conclusions?"

B) "Do you tend to spend or save disposable income?"

C) "Suppose you have just made a very good investment. How much of this successful decision do you tend to attribute to your own ability and to chance?"

D) "If you had some assets that have made a significant gain, but you decided to hold on to them because you expected they would increase in value and then they didn't, how would you feel about it?"

E) "If you were offered to choose among 6 free lottery tickets, would you choose the lottery ticket yourself or would you rather roll a 6-faced die?"

Self-attribution	A
Confirmation	B
Illusion of control	C
Self-control	D

Question 4

Which of the four behavioral types in the Pompian model is most likely to display endowment bias?

- Active accumulator
- Passive preserver
- Friendly follower
- Independent individualist

Question 5

What is the best description of Bowman's paradox?

- Bowman noticed that American households fail to take advantage of the higher premium equities allow over bonds. This is not in agreement with the fact that economic agents are supposed to be rational investors.
- Bowman noticed that investors have a preference for skewness despite the fact that rational investors should consider the negative consequences of skewness.
- Bowman stated that "good companies are usually bad stocks, bad companies are usually good stocks." This is in contrast to traditional finance's tenet that high risk should be associated with high returns.

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- Bowman noticed that riskiest firms provided the lowest returns. This is in contrast with traditional finance, which insists that higher risk should be associated with higher returns.

Question 6

Consider the following 4 lotteries:

Lottery 1: Win $5,500 with probability 1/5 or nothing.

Lottery 2: Win $3,000 with probability 1/3 or nothing.

Lottery 3: Win nothing with probability 2/3 or the right to participate in a lottery that pays $3,000 with probability 1.

Lottery 4: Win nothing with probability 2/3 or the right to participate in a lottery that pays $5,500 with probability 0.6.

Experimental studies have shown that most individuals choose Lottery 1 over Lottery 2. However, when confronted with Lottery 3 and Lottery 4, the majority chooses Lottery 3 over Lottery 4.

What behavioral bias can explain this?

- Framing bias
- Loss aversion
- Mental accounting
- Status quo

Question 7

During an interview with a client, you notice that she tends to make decisions regarding selecting a mutual fund or an online broker based on word of mouth or name recognition.

What kind of behavior bias do you think she is most likely displaying among those listed below?

- Availability
- Anchoring
- Representativeness
- Hindsight

Question 8

A **halo effect** is defined as extending a favorable evaluation of certain characteristics to other characteristics. For instance, a company with good growth record and good share price performance is seen as a good investment even going forward.

Which behavioral bias do you think is **most likely** behind the halo effect?

- availability
- hindsight

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- home bias
- representativeness

Question 9

Ben and Jerry are portfolio managers for one of the largest mutual funds in their town, which provides a variety of mutual funds for both corporate and individual clients. Ben has been a portfolio manager for over 10 years, and Jerry joined the firm 2 years ago after graduating from WQU. Ben is Jerry's supervisor.

During an afternoon meeting, Ben discusses the different strategies used at their firm. Summarizing the neglected firm strategy, he explains to Jerry that this strategy consists of looking for companies that are not covered by any analyst. The company does so because there is academic evidence that these firms can be good investments over a period of 3 years.

Is Ben showing traits of suffering from myopic loss aversion? Which of the following strategies best describes Ben's strategy? (**Note**: "best among" and not the "absolute best" because sometimes one has to take chances and choose the second or third best, another example of satisfice at work)

- A) support level
- B) resistance level
- C) moving average

- No. Moving average.
- Yes. Support level.
- Yes. Moving average.
- No. Support level.

Question 10

In what sense is "self-control" bias a problem for traditional finance (TF)?

- It is not a problem for traditional finance; it is a problem for behavioral finance.
- Investors who display self-control bias cannot be advised on how to construct their portfolios in an optimal manner.
- TF requires that individuals are able to save and invest in the early stages of life to fund their retirement.
- Self-control bias leads to unruly behaviors and could lead to investing too aggressively.

Question 11

Which of the following description best describes the multiple mental accounting version of BPT?

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- Assets are separated in layers based on risk, and investors integrate their portfolios into a single entity by considering covariance among the different layers.
- Each investor has a set of goals that addresses different needs that the investor tries to address using a multiple array of assets. Any combination(s) is good as long as we achieve our goals.
- Assets are separated in layers based on risk, and investors integrate their portfolios into a single entity using an approach similar to that used in mean variance optimization.
- Assets are separated in layers based on risk. The low-risk layers will contain bonds or riskless investments; the high risk level will consist of certain speculative equities, private equity investments, and derivatives.

Question 12

In the BB&K model, which confidence level and method of action underline the behavior of a guardian type of investor?

- Confident and impetuous, respectively.
- Anxious and careful, respectively.
- Confident and careful, respectively.
- Anxious and impetuous, respectively.

Question 13

In explaining the stock market participation puzzle, prospect theory uses myopic loss aversion. Which of these elements are more relevant in this case among:

A) functional forms of the value function

B) functional form of the weighting function

C) using the wrong time horizon to evaluate investment.

- A), B), and C)
- C)
- A)
- B)

Question 14

Consider that an economic agent faces the situation described in the table below.

Outcome	Value	probability	weight(probability)
x1	-50	0.03	2
x2	-5	0.05	1.5
x3	10	0.12	1
x4	80	0.80	.5

What is their utility in the context of prospect theory?

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- 29.83
- 63.45
- -57.50
- 0.66

Question 15

Steve Abenoza is a client of your firm. His previous advisor, John Morrison, CFA, has left the firm and you have been tasked with advising Mr. Abenoza.

Your first step is to go through the documents left behind by Mr. Morrison. From these you learn the following:

Your client is single, 59 years old, does not plan to marry or have children. He resides in New York City and has a second residence in Monte Carlo, as well.

Mr. Abenoza is a corporate lawyer with a prestigious NYC firm who draws a salary of $700,000 a year. His portfolio is worth approximately $5,000,000, a value reached primarily because of some successful investments in high risk oil and technology companies together with a generous number of stock options granted by Abenoza's employer.

Mr. Abenoza tends to live within his annual income (after tax), although he has been known to make a few extravagant expenses, beyond his annual income (net of taxes) every now and then.

A couple of years ago, Mr. Abenoza suffered from heart problems, but he has made a full recovery.

Mr. Abenoza intends to retire comfortably at age 65 with an income level of $200,000 per year and he plans to bequeath any assets remaining at his death to his alma mater, Yale University.

Mr. Morrison, his previous advisor, proposed that Mr. Abenoza follow a comprehensive financial plan. However, despite Morrison's recommendations, Mr. Abenoza has not changed his allocation and his portfolio has remained at nearly 80% percent equities, of which 35% consists of his employing company's stock (which is publicly traded)

You, and Mr. Morrison before you, believe that Mr. Abenoza should pursue a more conservative allocation with only 55% in equity, 35% in bonds, and 10% in cash instruments. You are concerned that a severe economic downturn could otherwise affect Mr. Abenoza's long-term financial plans as you fear that Mr. Abenoza could be tempted to sell his assets irrationally in the case of a downturn. Also, Mr. Morrison's calculation showed that Mr. Abenoza could achieve his financial goal with a more conservative allocation regardless.

During your first interview with Mr. Abenoza, you realize that Mr. Abenoza has not changed his portfolio allocations because he felt that the recommendations were too conservative. You also notice that he worries about missing out on hot stocks by not being aggressive enough.

Finally, you administer a risk tolerance questionnaire to Mr. Abenoza and then run a mean-variance optimization portfolio recommendation, which shows ideally a BMAA

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consisting of 65% equity, 25% bonds and 10% to cash instruments.

1. Is Mr. Abenoza displaying mainly emotional or cognitive biases?
2. Should you moderate or adapt?

- ◯ 1. Emotional biases.
  2. Adapt.
- ◯ 1. Cognitive biases.
  2. Moderate.
- ◯ 1. Cognitive biases.
  2. Adapt.
- ◯ 1. Emotional biases.
  2. Moderate.

Question 16

Nikhil Patel, a portfolio manager at M&M's Private Wealth Investment, is going over the information on one of his top clients:

- Giacomo Rossi is a 65 year old Italian billionaire with residence in Monte Carlo. He is married and has two children who are now adult and on their own.
- Mr. Rossi has accumulated his wealth through a successful business building machines for agriculture that he started nearly 40 years ago.
- Mr. Rossi is classified as cautious, and he is concerned about the future, in particular about protecting his assets as he is nearing retirement. His tolerance for risk is low.
- Mr. Rossi is willing to seek advice from those he perceives as more prepared than himself in matters of investing.
- Mr. Rossi's behavioral biases are listed as endowment, loss aversion, status quo.
- Mr. Rossi has indicated that, while indeed his risk tolerance is currently low, he would be willing to participate in reasonably risky investments if such investments had the the unanimous support of M&M's Private Wealth Investment Committee.

Which behavioral bias do you notice in the last pointer?

- ◯ Gambler's fallacy
- ◯ Confirmation bias
- ◯ Social proof
- ◯ Herding

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